

Faculty of Engineering and Technology

B.Tech (CS&E)

MOOC2 – Core Python

1: Write a Python program to:

- Accept a number from the user
- Display its absolute value, square, and cube
- Round it to 2 decimal places

```
num = float(input("Enter a number: "))
print("Absolute value:", abs(num))
print("Square:", pow(num, 2))
print("Cube:", pow(num, 3))
print("Rounded (2 decimal places):", round(num, 2))
```

2: Write a Python program to:

- Compute square root, floor, and ceiling of a number
- Calculate log base 10 of a number

```
import math

num = float(input("Enter a number: "))
print("Square root:", math.sqrt(num))
print("Floor value:", math.floor(num))
print("Ceiling value:", math.ceil(num))
print("Log base 10:", math.log10(num))
```

3: Write a program to:

- Take an angle in degrees as input
- Convert it to radians
- Find its sine, cosine, and tangent

```
import math
deg = float(input("Enter angle in degrees: "))
rad = math.radians(deg)
print("Sine:", math.sin(rad))
```

```
print("Cosine:", math.cos(rad))
print("Tangent:", math.tan(rad))
```

4: Write a program to:

- Calculate the factorial of a number
- Find the greatest common divisor (GCD) of two numbers

```
import math
num = int(input("Enter a number for factorial: "))
print("Factorial:", math.factorial(num))

a = int(input("Enter first number for GCD: "))
b = int(input("Enter second number: "))
print("GCD:", math.gcd(a, b))
```

5: Write a program that:

- Generates a random number between 1 and 100

```
import random
print("Random number (1 to 100):", random.randint(1, 100))
```

Trivia Time :

1. Create a simple Python program that uses math functions to create a fun number-guessing game. This game generates a random number and asks the player to guess it. It uses `math.log2()` to tell the player how many guesses they should take ideally.

1. Objective:
Guess the secret number chosen randomly by the computer within a range of 1 and 100 (inclusive).
2. Guess Limit:
You are encouraged to guess the number in no more than $\log_2(N)$ attempts, where N is the size of the range. For example, for 1–100, this means about 7 guesses.
3. Feedback After Each Guess:
 - If your guess is too low, you'll be told to guess higher.
 - If your guess is too high, you'll be told to guess lower.
 - If your guess is correct, you win!
4. Input Rules:
 - Enter only whole numbers (integers) within the specified range. Invalid input will prompt a warning, and the game will ask you to try again.